Multi-level Asymmetric Contrastive Learning for Medical Image Segmentation Pre-training

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***Abstract*— Medical image segmentation is a fundamental yet challenging task due to the arduous process of acquiring large volumes of high-quality labeled data from experts. Contrastive learning offers a promising but still problematic solution to this dilemma. Firstly existing medical contrastive learning strategies focus on extracting image-level representation, which ignores abundant multi-level representations. Furthermore they underutilize the decoder either by random initialization or separate pre-training from the encoder, thereby neglecting the potential collaboration between the encoder and decoder. To address these issues, we propose a novel multi-level asymmetric contrastive learning framework named MACL for enhancing medical image segmentation. Specifically, we design an asymmetric contrastive learning structure to pre-train encoder and decoder simultaneously to provide better initialization for segmentation models. Moreover, we develop a multi-level contrastive learning strategy that integrates correspondences across feature-level, image-level, and pixel-level representations to ensure the encoder and decoder capture comprehensive details from representations of varying scales and granularities during the pre-training phase. Finally, experiments on 8 medical image datasets indicate our MACL framework outperforms existing 11 contrastive learning strategies. *i.e.* Our MACL achieves a superior performance with more precise predictions from visualization figures and 1.72%, 7.87%, 2.49% and 1.48% Dice higher than previous best results on ACDC, MMWHS, HVSMR and CHAOS with 10% labeled data, respectively. And our MACL also has a strong generalization ability among 5 variant U-Net backbones. Our code will be released after acceptance [1].**

***Index Terms*— Medical Image Segmentation, Self-supervised Learning, Contrastive Learning**

I. Introduction

MEDICAL image segmentation [1]–[3], which involves partitioning images into anatomical structures or pathological regions, is fundamental to computer-aided diagnosis.

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[1]code ocean link: `https://doi.org/10.24433/CO.1298605.v1`

Although models with fully supervised training have achieved remarkable success in this domain, they still rely heavily on large amounts of expert-annotated data which is costly, time-consuming and expertise-required. Meanwhile, large volumes of unlabeled medical images (*e.g.* CT and MRI) are generated everyday worldwide, highlighting the need for methods that can leverage such numerous unlabeled data for pre-training and achieve high performance with limited annotations.

Self-supervised learning (SSL) has emerged as a powerful approach to mitigate the reliance on large annotated datasets by learning meaningful representations from unlabeled data. Among various SSL methods, contrastive learning (CL) has achieved remarkable success in both natural and medical image analysis by learning discriminative image-level features through a contrastive loss that brings similar representations (*a.k.a.* positive pairs) closer and pushes dissimilar representations (*a.k.a.* negative pairs) apart in the feature space.

In medical image segmentation, several medical CL frameworks [4]–[8] in Figure 1(a) and (b) have shown exciting results, which inspire us to excavate the potential of medical CL. They typically pre-train a U-Net encoder using image-level contrastive loss to extract inter-image discrimination from unlabeled upstream datasets. And then they combine the well pre-trained encoder with a randomly initialized decoder for fine-tuning on downstream datasets with a small fraction of labeled data. However, these approaches exhibit two key limitations: (1) Medical image segmentation aims to predict the class label for each pixel within one image, which focuses on intra-image discrimination, instead of inter-image discrimination. In this sense, it poses demands for more distinctive pixel-level representation learning, which will be more appropriate for medical image segmentation. However, current works [4]–[8] focus on extracting image-level projection output from the encoder along with an image-level projector, which do not exploit the details of both feature-level projection from the encoder and pixel-level projection from the decoder. Therefore, these pre-trained models can be sub-optimal for segmentation task due to the discrepancy between image-level projection and pixel-level projection and the missing of the implicit feature-level consistency. (2) The symmetric encoder-decoder architecture is widely used for fully supervised medical image segmentation due to its capacity to extract multi-level representations (encoder) and preserve spatial details (decoder and skip connection). Such an efficient collaborative work paradigm inspires us that incorporating a decoder together with an encoder for CL pre-training should enhance the performance of downstream segmentation task. However, those proposed medical CL frameworks do not make full use of the decoder because their decoder is either randomly initialized and trained during downstream fine-tuning [4]–[7] or pre-trained separately from the encoder, ignoring the collaboration between the encoder and decoder during pre-training [8]. Specifically, the feature map from the encoder is highly compressed with fewer details which is suitable for global image-level CL while the feature map from the decoder is fine-grained with more texture which aligns well with local pixel-level CL. That means pre-training encoder and decoder simultaneously can allow the model to capture features at different levels and complement each other.

To fill the gap, we propose a novel **M**ulti-level **A**symmetric **CL** framework named **MACL** shown in Figure 1(c) by introducing an asymmetric CL structure and a multi-level CL strategy to realize one-stage encoder-decoder synchronous pre-training. Different from the existing works, there are mainly two characteristics of our proposed framework: 1) We propose a multi-level CL strategy (*i.e. image-level, feature-level, pixel-level*) from global to local to make sure luxuriant details from multi-scale and multi-granularity representations can be learned by the encoder and decoder during pre-training. 2) We incorporate a decoder into normal *one-encoder* CL framework to realize the synchronous encoder-decoder pre-training and further make an adaptive modification to the decoder which finally forms an asymmetric CL structure. Specifically, instead of directly introducing a symmetric encoder-decoder structure to both of the two branches, we introduce an encoder with a partial decoder in the dominant branch while only an encoder in the auxiliary branch. With such an asymmetric structure, our MACL can delightfully reduce the computation complexity of the model and guarantee sufficient negative pairs for CL. According to our experimental results, our proposed MACL outperforms other 11 methods across 8 medical image datasets.

II. Related Work

A. Medical Image Segmentation

In recent years, deep learning—particularly convolutional neural networks (CNNs) – has achieved remarkable success in medical image segmentation [9]–[15]. Foundational architectures such as FCN, U-Net, and DeepLab serve as key milestones. U-Net [9], with its encoder-decoder structure and skip connections, enables accurate delineation of anatomical features. Building upon U-Net's success, attention U-Net [10] incorporates attention gates (AGs) to the skip connections to implicitly learn to suppress irrelevant regions in the input image while highlighting the regions of interest for the segmentation task. Similarly, in order to bridge the semantic gaps for an accurate medical image segmentation, UCTransNet [11] proposes a CTrans module to replace the original skip connection. This module consists of two sub-modules: CCT, which conducts multi-scale channel cross-fusion with a transformer, and CCA, which incorporates channel-wise cross-attention. Moreover, RollingUNet [12] combines MLP with U-Net to efficiently fuse local features and long-range dependencies. Additionally, the development of Kolmogorov-Arnold Networks (KANs) [16] offered superior interpretability and efficiency through the utilization of a series of nonlinear, learnable activation functions. Therefore, UKAN [13] integrated KANs into UNet framework, augmenting its capacity for non-linear modeling while also improving model interpretability.

B. Contrastive Learning for Enhancing Medical Image Segmentation

Contrastive learning (CL), a variant of self-supervised learning (SSL), has demonstrated strong capability in extracting image-level features from large-scale unlabeled data, significantly reducing annotation costs. CL plays a vital role in tasks such as video frame prediction, semantic segmentation, and object detection by attracting positive sample pairs and repulsing negative ones using the InfoNCE loss. Early methods like MoCo [17] and SimCLR [18] emphasize the importance of abundant negative samples, achieved via a memory bank or large batch sizes. Later approaches such as SwAV [19], BYOL [20], and SimSiam [21] demonstrate that effective representations can also be learned without explicit negatives, using techniques like clustering, momentum encoders, and stop-gradient.

In the medical image segmentation domain, substantial efforts of contrastive learning [4]–[8] have been devoted to incorporating unlabeled data to improve network performance due to the limited data and annotations. There are mainly two categories in medical CL: design of siamese networks and construction of contrastive pairs. In terms of the design of siamese networks, DeSD [5] addresses weak supervision in shallow layers via deep self-distillation, while DiRA [6] combines discriminative, restorative, and adversarial learning for richer representations. For constructing contrastive pairs, SimTriplet [7] uses triplet-based positive samples to exploit multi-view characteristics, and GCL [8] employs *partition-based* strategies to form local contrastive loss for volumetric data. PCL [4] further leverages positional information to mitigate false negatives, improving segmentation quality in cases where structural similarities across images pose challenges. Recently, SuperCL [22] introduces superpixel-guided contrastive pairs generation, featuring Intra-image Local Contrastive Pairs (ILCP) and Inter-image Global Contrastive Pairs (IGCP) by utilizing superpixel maps as pseudo masks.

A primary shortcoming of current CL frameworks, including SuperCL, is the neglect of decoder pre-training. Methods like DeSD, DiRA, and SuperCL focus solely on encoder optimization, leaving the decoder randomly initialized during fine-tuning. This creates a misalignment between pre-training and segmentation objectives, compromising feature learning for precise localization. Furthermore, these methods often operate at a single representation level (global or local), overlooking multi-scale feature integration, and incur inefficiencies through multi-stage designs or external dependencies. These limitations motivate our Multi-level Asymmetric Contrastive Learning (MACL) framework, which addresses gaps via synchronous encoder-decoder pre-training in an asymmetric one-stage structure. By integrating feature-level, image-level, and

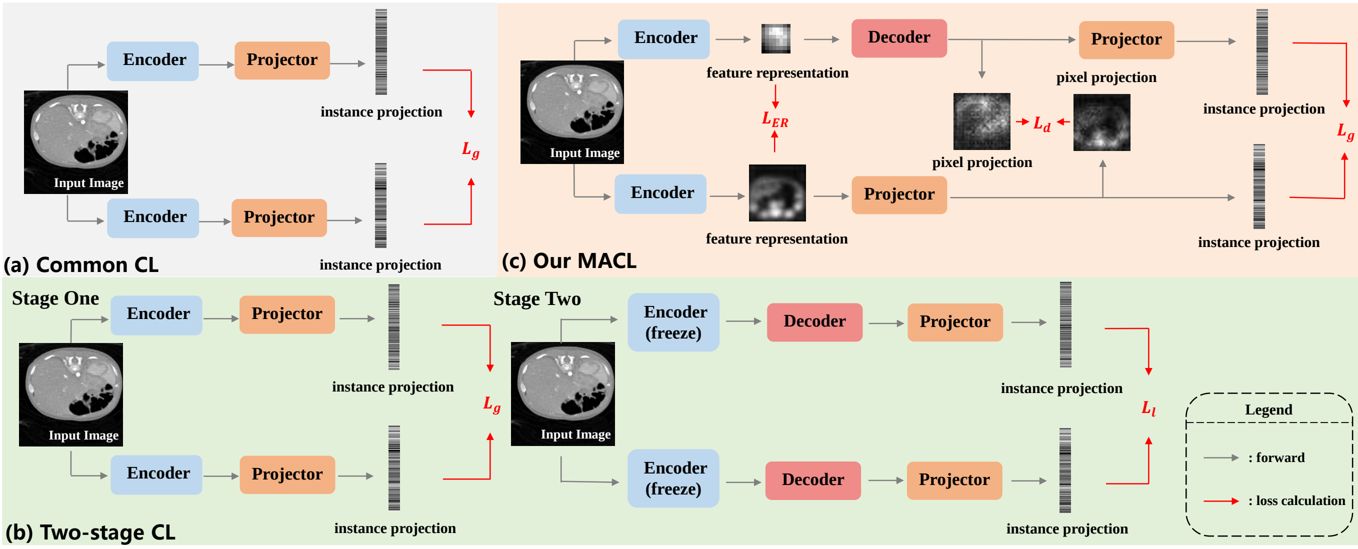

Fig. 1. Comparison of different CL frameworks for medical images: (a) Common CL frameworks used for medical image segmentation are symmetric and similar to SimCLR with a global contrastive loss $\mathcal{L}_g$ for optimization. (b) Two-stage CL framework pre-trains the encoder with global contrastive loss $\mathcal{L}_g$ and decoder with local contrastive loss $\mathcal{L}_l$ in separate stage which ignores the collaboration between the encoder and decoder. (c) Our proposed MACL framework is asymmetric with an additional partial decoder for pre-training and integrates multi-level CL strategy including feature-level equivariant regularization, image-level and pixel-level CL to get better initialization of both encoder and decoder for downstream segmentation tasks.

pixel-level contrasts, MACL captures granular details without external mechanisms like superpixels, ensuring efficient and holistic representation learning.

III. Methodology

A. Problem Definition

Given an input image $\boldsymbol{X} \in \mathbb{R}^{C_x \times H_x \times W_x}$, medical image segmentation aims to classify each pixel with a segmentation network, where $H_x \times W_x$ is the resolution, C_x is the channel of the input image. To achieve this purpose, the network needs an encoder $e(\cdot)$, parameterized by θ_e, to extract multi-level features and a decoder $d(\cdot)$, parameterized by θ_d, to fuse the features into $\boldsymbol{Z}$ to recover image details:

$$\boldsymbol{Z} = d(e(\boldsymbol{X})) = d(\{\boldsymbol{X}^1, \cdots, \boldsymbol{X}^L\}) \tag{1}$$

where $\boldsymbol{X}^i$ represents the i_{th}-level feature. L is the number of encoder layers. This fused feature $\boldsymbol{Z}$ is finally classified by a segmentation head to output segmentation map.

Except for optimizing $e(\cdot)$ and $d(\cdot)$ from scratch, CL focuses on pre-training them with numerous unlabeled data to provide suitable initialization before supervised learning for downstream segmentation task. However, as shown in Figure 1(a), most existing CL frameworks [4]–[8] used for medical image segmentation are symmetric and similar to SimCLR [18], which only pre-trains an encoder $e(\cdot)$ with a global contrastive loss $\mathcal{L}_g$ among image-level sample pairs:

$$\arg\min_{\theta_e} \quad \mathcal{L}_g(e(\tilde{\boldsymbol{X}}), e(\hat{\boldsymbol{X}})) \tag{2}$$

where $\tilde{\boldsymbol{X}}$ and $\hat{\boldsymbol{X}}$ are two random augmentations of the input image $\boldsymbol{X}$. The decoder structure $d(\cdot)$ is not considered in the above pre-training process, which ignores the importance of decoders to the downstream segmentation task. To solve this problem, as shown in Figure 1(b), GCL [8] proposes a two-stage CL framework which utilizes two pre-training stages to optimize $e(\cdot)$ with a global contrastive loss $\mathcal{L}_g$ and $d(\cdot)$ with a local contrastive loss $\mathcal{L}_l$ respectively:

$$\begin{cases} \arg\min\limits_{\theta_e} \quad \mathcal{L}_g(e(\tilde{\boldsymbol{X}}), e(\hat{\boldsymbol{X}})), \quad \textit{in stage one} \\ \arg\min\limits_{\theta_d} \quad \mathcal{L}_l(d(e(\tilde{\boldsymbol{X}})), d(e(\hat{\boldsymbol{X}}))), \quad \textit{in stage two} \end{cases} \tag{3}$$

The parameters of $e(\cdot)$ are freezed in stage two. Although this strategy takes the decoder $d(\cdot)$ into consideration, the two-stage training process ignores the collaboration between the encoder and decoder during pre-training, which makes the decoder not able to learn image-level representation well due to the absence of global contrastive loss.

Based on the above issues, in order to realize the synchronous training of $e(\cdot)$ and $d(\cdot)$ to learn multi-level representations, we propose a novel one-stage multi-level asymmetric CL framework shown in Figure 1(c). Our MACL framework pre-trains the encoder $e(\cdot)$ and decoder $d(\cdot)$ simultaneously with both image-level global contrastive loss $\mathcal{L}_g$ and pixel-level dense contrastive loss $\mathcal{L}_d$ to learn multi-level feature representations:

$$\arg\min_{\theta_e, \theta_d} \quad \mathcal{L}_g(d(e(\tilde{\boldsymbol{X}})), d(e(\hat{\boldsymbol{X}}))) + \mathcal{L}_d(d(e(\tilde{\boldsymbol{X}})), d(e(\hat{\boldsymbol{X}}))) \tag{4}$$

With our MACL, both encoder and decoder can synchronously learn multi-level representations and get better initialization.

B. Asymmetric Contrastive Learning Structure

In order to realize the one-stage synchronous training of encoder $e(\cdot)$ and decoder $d(\cdot)$ for better assisting the downstream

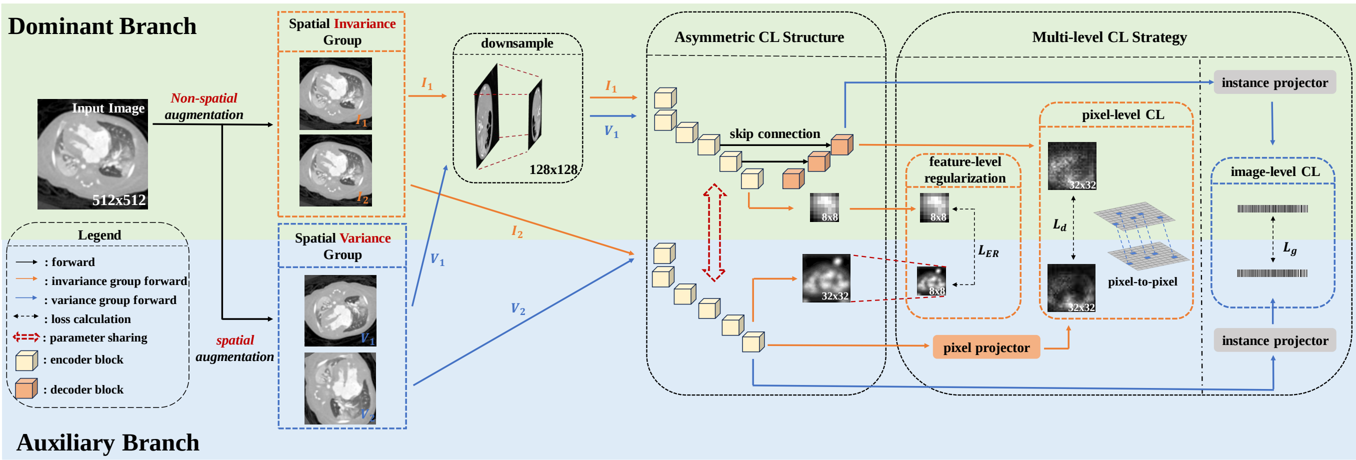

Fig. 2. Overview of our proposed MACL framework. The input image is propagated to two branches: the dominant branch and auxiliary branch. In the dominant branch, after augmentation and downsampling, the input image is fitted into the encoder to get feature-level representation, and then propagated to the decoder and projector to get image-level and pixel-level projections; while in the auxiliary branch, it is the same process but no decoder will be used. The feature-level representations from two branches will be used in equivariant regularization loss, while image-level and pixel-level projections will be used in global and dense contrastive loss, respectively to achieve multi-level contrastive learning.

segmentation tasks, our MACL elaborates an asymmetric CL structure as shown in Figure 2. This proposed structure contains two branches: (1) *dominant branch* which pre-trains both encoder and decoder simultaneously to get better initialization for downstream segmentation tasks and (2) *auxiliary branch* which serves as a contrastive counterpart to assist the dominant branch for pre-training.

Before being fed into the asymmetric CL structure, two different settings for data augmentations will be applied to the input image $\boldsymbol{X}$: (1) One set is only non-spatial augmentations T_{fix} which will be used for CL involving pixel-wise comparison including feature-level and pixel-level CL; (2) The other set is the combination of non-spatial and spatial augmentations T_{var} which is suitable for image-level CL not involving pixel-wise comparison:

$$\{\boldsymbol{I_1}, \boldsymbol{I_2}\} = T_{fix}(\boldsymbol{X}), \quad \{\boldsymbol{V_1}, \boldsymbol{V_2}\} = T_{var}(\boldsymbol{X}) \tag{5}$$

where $\boldsymbol{I_1}, \boldsymbol{I_2}$ are augmented images of spatial invariance group and $\boldsymbol{V_1}, \boldsymbol{V_2}$ are augmented images of spatial variance group.

1) Auxiliary Branch: The auxiliary branch keeps in line with that of common CL strategies: an encoder $e(\cdot)$ followed by a shallow multi-layer perceptron (MLP) projection head $\tilde{g}(\cdot)$ [18] (but including an image-level projector $\tilde{g}_{ins}(\cdot)$ and pixel-level projector $\tilde{g}_{pix}(\cdot)$) and no decoder. Specifically, the augmented image with pixel position fixed $\boldsymbol{I_2}$ is propagated to the encoder $e(\cdot)$ to get feature-level representation $\tilde{\boldsymbol{Y}}$ and then propagated into the pixel-level projector $\tilde{g}_{pix}(\cdot)$ to get pixel-level projection $\tilde{\boldsymbol{Z}}^l \in \mathbb{R}^{B\times C\times H\times W}$ (where B denotes batch size, C denotes channels of the projection, $H \times W$ denotes the spatial size of the projection). While the augmented image with pixel position changed $\boldsymbol{V_2}$ is propagated into the encoder $e(\cdot)$ and the image-level projector $\tilde{g}_{ins}(\cdot)$ to get image-level projection $\tilde{\boldsymbol{Z}}^g \in \mathbb{R}^{B\times C}$:

$$\tilde{\boldsymbol{Y}} = e((\boldsymbol{I_2})), \ \tilde{\boldsymbol{Z}}^l = \tilde{g}_{pix}(\tilde{\boldsymbol{Y}}), \ \tilde{\boldsymbol{Z}}^g = \tilde{g}_{ins}(e(\boldsymbol{V}_2)) \tag{6}$$

2) Dominant Branch: Unlike the auxiliary branch, the dominant branch introduces a decoder $d(\cdot)$ after the encoder $e(\cdot)$ for synchronous pre-training to extract richer feature-level information for downstream segmentation. This branch includes both image-level and pixel-level MLP projection heads $g_{ins}(\cdot)$ and $g_{pix}(\cdot)$. Moreover, to address the mismatch in feature map sizes caused by the decoder introduction in our proposed asymmetric CL structure, a downsampling operation $down(\cdot)$ is added during data augmentation, ensuring alignment and maintaining sufficient negative pairs with low computational cost. Specifically, the pixel-position-fixed augmented image $\boldsymbol{I_1}$ is first downsampled and encoded to get the feature-level representation $\boldsymbol{Y}$, and then decoded to get the pixel-level projection $\boldsymbol{Z}^l$, while the pixel-position-changed augmented image $\boldsymbol{V_1}$ is downsampled, encoded, and passed through $g_{ins}(\cdot)$ to get image-level projection $\boldsymbol{Z}^g$:

$$\boldsymbol{Y} = e(down(\boldsymbol{I}_1)), \ \boldsymbol{Z}^l = d(\boldsymbol{Y}), \ \boldsymbol{Z}^g = g_{ins}(e(down(\boldsymbol{V}_1))) \tag{7}$$

C. Multi-level Contrastive Learning Strategy

Under the guarantee of our proposed asymmetric CL structure in Section B which pre-trains the encoder and decoder simultaneously, we can optimize our MACL framework with multi-level contrastive learning strategy considering image-level, pixel-level and feature-level representations as follows:

$$\mathcal{L}_{MLC} = \lambda_1\mathcal{L}_g + \lambda_2\mathcal{L}_d + \lambda_3\mathcal{L}_{ER} \tag{8}$$

where $\lambda_1, \lambda_2, \lambda_3$ are the weighting factors, $\mathcal{L}_{MLC}$ is the total multi-level contrastive loss, and $\mathcal{L}_g, \mathcal{L}_d, \mathcal{L}_{ER}$ will be discussed later. The overall pre-training is conducted in one-stage manner.

1) Global Contrastive Learning: To enhance image-level representation, we adopt a global contrastive loss. Given a set of N slices $\{X_i\}_{i=1\ldots N}$, each image undergoes two random augmentations, resulting in a mini-batch of $2N$ samples $\left\{\tilde{X}_i\right\}_{i=1\ldots 2N}$, where $\tilde{X}_{2i}$ and $\tilde{X}_{2i-1}$ are augmented views of

TABLE I
QUANTITATIVE RESULTS OF OUR MACL AGAINST OTHER SOTA METHODS ON 4 MULTI-ORGAN SEGMENTATION DATASETS.

Labeled	Methods	ACDC (100 patients)				MMWHS (20 patients)				HVSMR (10 patients)				CHAOS (20 patients)			
		DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓
10 %	Scratch	79.99	67.39	8.78	3.82	63.04	48.89	9.31	3.39	72.81	59.02	34.34	10.18	51.76	37.51	23.08	9.90
	SimCLR [18]	78.60	65.47	8.11	3.36	61.70	46.59	9.67	3.53	75.78	62.34	25.16	**6.22**	53.22	38.11	22.28	9.04
	MoCo [17]	81.34	69.13	8.12	3.20	57.54	42.62	11.33	4.15	76.20	62.79	27.45	8.33	53.14	38.49	21.45	9.53
	BYOL [20]	77.55	64.17	9.05	3.96	63.75	49.54	9.34	3.61	73.57	59.87	29.79	8.51	50.45	35.66	24.20	10.32
	SwAV [19]	82.11	70.20	6.35	2.46	64.27	49.96	10.31	3.98	76.28	62.91	24.09	6.42	49.87	35.04	19.75	8.86
	SimSiam [21]	71.96	57.01	11.65	4.94	61.37	46.51	10.61	3.89	75.10	61.47	29.73	8.72	41.21	28.87	21.55	10.20
	SimTriplet [7]	71.07	56.07	11.31	5.20	62.52	47.81	10.98	4.10	72.66	58.48	30.72	8.65	32.88	23.14	27.81	12.90
	GCL [8]	84.86	74.07	7.07	2.85	71.41	58.58	7.27	2.55	75.70	62.22	28.15	7.68	58.78	44.00	**15.28**	6.36
	PCL [4]	84.91	74.16	6.54	2.65	65.98	52.32	8.07	3.16	76.12	62.82	24.50	6.43	63.61	48.23	16.15	6.79
	DeSD [5]	76.02	62.27	10.25	4.15	57.03	42.30	10.89	4.36	74.57	61.16	24.00	6.40	61.93	46.68	18.91	8.39
	DiRA [6]	83.14	71.61	6.73	2.60	57.73	42.95	10.92	3.76	74.87	61.35	27.91	8.01	44.30	31.24	24.63	11.93
	MACL (Ours)	**86.63**	**76.72**	**5.23**	**1.73**	**79.28**	**67.19**	**5.91**	**1.98**	**78.77**	**66.05**	**22.38**	6.25	**65.09**	**51.12**	18.95	7.40
25 %	Scratch	88.99	80.34	3.93	1.24	85.14	74.87	4.66	1.45	81.26	69.25	20.63	5.85	66.71	52.74	12.32	5.39
	SimCLR [18]	88.55	79.71	4.53	1.47	86.14	76.03	4.59	1.36	80.53	68.50	23.42	6.75	66.15	52.21	11.76	4.75
	MoCo [17]	88.38	79.39	5.01	1.55	82.89	71.46	5.40	1.69	81.26	69.18	20.41	5.64	65.33	50.93	15.12	6.40
	BYOL [20]	89.27	80.78	4.62	1.64	87.63	78.43	3.83	1.20	81.33	69.30	19.95	5.80	62.49	48.31	14.13	5.71
	SwAV [19]	88.99	80.34	4.90	1.78	87.71	78.61	3.78	1.22	81.30	69.18	17.24	4.62	68.12	54.01	12.71	5.15
	SimSiam [21]	88.24	79.18	4.85	1.71	82.71	71.19	5.25	1.78	79.70	67.30	22.83	6.31	51.82	39.11	14.82	6.86
	SimTriplet [7]	87.99	78.80	5.78	2.00	82.43	70.82	5.24	1.73	80.51	68.24	24.34	7.17	63.55	49.30	13.23	5.98
	GCL [8]	88.55	79.67	4.22	1.34	88.01	79.04	4.02	1.17	82.41	70.80	22.08	6.59	76.88	63.48	9.11	3.70
	PCL [4]	88.97	80.33	4.17	1.43	86.65	76.90	4.23	1.28	83.13	71.70	20.28	5.17	78.98	66.08	8.87	3.32
	DeSD [5]	87.14	77.47	6.11	2.19	84.46	73.66	5.15	1.70	81.44	69.44	21.19	5.57	67.96	54.28	11.70	4.90
	DiRA [6]	88.92	80.22	4.24	1.48	86.68	77.08	4.46	1.43	81.56	69.67	20.85	5.69	64.00	49.87	13.47	5.69
	MACL (Ours)	**89.64**	**81.40**	**3.59**	**1.20**	**89.90**	**81.92**	**3.68**	**1.01**	**83.68**	**72.47**	**17.03**	**4.14**	**79.88**	**67.45**	**8.07**	**3.05**
100 %	Scratch	92.07	85.38	3.06	1.07	92.27	85.79	2.71	0.76	84.95	74.33	14.83	3.56	88.65	79.84	4.38	1.62

X_i. The projections from the dominant and auxiliary branches are denoted as Z_i^g, $\tilde{Z}_i^g$, respectively. The global contrastive loss is defined as:

$$\mathcal{L}_g = \sum_{i=1}^{2N} -\frac{1}{\left|\Omega_i^+\right|} \sum_{j \in \Omega_i^+} \log \frac{e^{sim\left(Z_i^g, \tilde{Z}_j^g\right)/\tau}}{\sum_{k=1}^{2N} \mathbb{I}_{i \neq k} \cdot e^{sim\left(Z_i^g, \tilde{Z}_k^g\right)/\tau}} \tag{9}$$

where $\left|\Omega_i^+\right|$ is the cardinality of the position-based positive indices set regarding Z_i^g, $sim(\cdot,\cdot)$ is the cosine similarity, τ is the temperature and $\mathbb{I}$ is the indicator function. Unlike standard contrastive loss [18], which uses a single positive pair, we follow PCL [4] to generate multiple position-aware contrastive pairs.

2) Dense Contrastive Learning: To capture fine-grained pixel-level semantics, we propose a dense contrastive loss. For each augmented image $\tilde{X}_i$, pixel-level projections Z_i^l and $\tilde{Z}_i^l$ are generated by the pixel-level projector in the dominant and auxiliary branch, respectively, which are dense feature maps with a size of $H/2^L \times W/2^L$. Each pair of augmented images yields $H/2^L \times W/2^L$ pixel-level projection pairs. Positive and negative pairs are constructed using position-based strategy, allowing the extension of global contrastive learning to a dense, pixel-level setting:

$$\mathcal{L}_d = \sum_{i=1}^{2N} -\frac{1}{\left|\Omega_i^+\right|} \sum_{j \in \Omega_i^+} \frac{1}{\frac{H \times W}{2^{2L}}} \sum_s \log \frac{e^{sim\left(Z_{i,s}^l, \tilde{Z}_{j,s}^l\right)/\tau}}{\sum_{k=1}^{2N} \mathbb{I}_{i \neq k} \cdot e^{sim\left(Z_{i,s}^l, \tilde{Z}_{k,s}^l\right)/\tau}} \tag{10}$$

where $Z_{i,s}^l$ and $\tilde{Z}_{i,s}^l$ denote the s_{th} output of $H/2^L \times W/2^L$ pixel-level projection pairs of $\tilde{X}_i$ from the dominant branch and auxiliary branch, respectively.

3) Equivariant Regularization: The feature-level representation is governed by equivariant regularization. While downsampling successfully resolves the feature size mismatch for pixel-level learning, it introduces a secondary, subtler issue related to feature-level consistency. The representations extracted by the encoders in the two branches are now derived from inputs at different resolutions, hence introducing an implicit feature-level discrepancy. To mitigate any potential inconsistency arising from this downsampling step and to enforce a robust feature representation that is invariant to such input variations, we introduce equivariant regularization into our multi-level CL to apply consistency regularization on multi-scale feature representations to provide further self-supervision for network learning. The equivariant regularization can be formulated as:

$$\mathcal{L}_{ER} = \|\boldsymbol{Y} - down(\tilde{\boldsymbol{Y}})\|_1 \tag{11}$$

where $\boldsymbol{Y}$ and $\tilde{\boldsymbol{Y}}$ are the representations learned from the encoder in the dominant and auxiliary branch, respectively. We use L1 norm here to integrate consistency regularization on feature representations from the encoders to provide further self-supervision for network learning.

IV. EXPERIMENTS

A. Datasets

We evaluate our proposed MACL on two segmentation tasks: multi-organ segmentation and region-of-interest (ROI)-based segmentation, using 3 public datasets for pre-training (CHD, BraTS2018, KiTS2019) and 8 for fine-tuning.

Multi-organ Segmentation: **CHD dataset** [23] (17525 CT slices) includes 68 3D cardiac images with annotations for seven heart substructures and used for multi-organ CT segmentation pre-training. **ACDC dataset** [24] provides 3D cardiac MRIs from 100 patients, with labels for LV, RV, and myocardium. **MMWHS dataset** [25] includes 20 annotated cardiac CT scans used for fine-tuning. **HVSMR dataset** [26] offers 10 3D cardiac MRIs labeled for blood pool and myocardium. **CHAOS dataset** [27] contains 120 DICOM

TABLE II
QUANTITATIVE RESULTS OF OUR PROPOSED MACL AGAINST OTHER SOTA METHODS ON 4 ROI-BASED SEGMENTATION DATASETS.

Labeled	Methods	Spleen (41 patients)				ISIC (2594 images)				Heart (20 patients)				Hippocampus (260 patients)			
		DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓
10 %	Scratch	66.63	49.95	12.58	5.78	<u>87.21</u>	<u>77.32</u>	23.47	9.72	73.30	57.85	5.21	1.82	<u>78.27</u>	<u>64.30</u>	1.36	0.51
	SimCLR [18]	60.38	43.24	12.42	4.67	85.85	75.21	23.81	10.32	73.42	58.00	**4.32**	<u>1.40</u>	75.91	61.18	1.48	0.54
	MoCo [17]	56.49	39.36	21.32	9.35	85.61	74.84	24.56	9.95	68.32	51.88	6.71	2.48	78.27	64.30	**1.27**	0.46
	BYOL [20]	61.38	44.28	14.56	6.32	87.00	77.00	24.20	10.28	70.34	58.00	6.17	2.16	77.24	62.92	1.39	0.49
	SwAV [19]	61.87	44.79	13.61	6.53	85.85	75.20	24.19	10.45	72.31	56.63	5.95	2.27	76.00	61.29	1.47	0.52
	SimSiam [21]	67.88	51.37	12.89	3.65	83.02	70.97	30.04	13.70	58.56	41.41	10.71	5.15	73.45	58.05	1.39	0.49
	SimTriplet [7]	63.56	46.59	16.95	7.54	83.01	70.96	29.27	13.03	57.76	40.61	8.85	3.94	74.80	59.74	1.31	<u>0.46</u>
	GCL [8]	<u>73.40</u>	<u>57.97</u>	11.79	4.99	85.53	74.72	24.37	10.84	76.88	62.44	5.21	1.73	78.14	64.13	1.32	0.46
	PCL [4]	71.22	55.30	11.10	4.97	86.64	76.43	24.41	10.50	<u>78.47</u>	<u>64.57</u>	5.95	2.14	78.13	64.11	<u>1.29</u>	0.47
	DeSD [5]	64.24	47.31	<u>7.18</u>	**2.83**	83.56	71.76	25.66	11.50	77.55	63.33	5.80	2.28	74.48	59.34	1.64	0.62
	DiRA [6]	66.23	49.51	20.96	8.93	87.02	77.02	<u>22.91</u>	<u>9.28</u>	70.19	54.07	<u>4.76</u>	**1.36**	77.15	62.80	1.36	0.50
	MACL (Ours)	**80.93**	**67.97**	**6.93**	<u>2.92</u>	**87.45**	**77.70**	**21.53**	**8.91**	**79.37**	**65.79**	5.40	1.99	**78.33**	**64.38**	1.31	**0.46**
25 %	Scratch	74.99	59.98	9.14	4.33	88.44	79.28	**16.32**	**6.41**	78.35	64.41	4.57	1.33	83.48	71.65	1.08	0.35
	SimCLR [18]	81.75	69.13	7.05	3.06	87.82	78.28	17.79	7.18	80.07	66.76	**3.48**	0.98	83.35	71.46	1.05	0.36
	MoCo [17]	75.98	61.26	8.55	4.09	87.53	77.83	17.87	7.01	75.35	60.45	5.60	2.11	83.94	72.33	<u>0.98</u>	<u>0.33</u>
	BYOL [20]	75.15	60.19	6.34	2.39	88.50	79.38	<u>16.78</u>	<u>6.54</u>	80.70	67.65	4.36	1.16	83.38	71.50	1.04	0.35
	SwAV [19]	80.40	67.23	<u>5.52</u>	<u>2.34</u>	87.90	78.41	17.65	6.77	81.94	69.40	3.98	1.22	83.05	71.02	1.06	0.35
	SimSiam [21]	78.70	64.88	8.87	4.56	87.29	77.45	19.31	7.52	71.65	55.83	<u>3.74</u>	1.17	80.60	67.51	1.11	0.37
	SimTriplet [7]	78.81	65.04	8.84	3.96	86.52	76.24	19.50	7.68	76.42	61.83	5.11	1.72	82.01	69.51	1.06	0.35
	GCL [8]	<u>84.37</u>	<u>72.97</u>	6.25	2.52	<u>88.52</u>	<u>79.41</u>	17.55	7.06	82.30	69.92	3.76	<u>0.96</u>	83.63	71.87	1.07	0.37
	PCL [4]	83.55	71.74	7.03	2.96	88.26	78.99	18.04	7.27	<u>82.69</u>	<u>70.49</u>	3.78	**0.93**	<u>83.98</u>	<u>72.40</u>	**0.96**	0.33
	DeSD [5]	81.02	68.10	6.59	3.28	87.34	77.52	20.29	8.41	81.50	68.77	4.45	1.43	81.81	69.22	1.08	0.37
	DiRA [6]	80.80	67.78	7.95	4.17	88.38	79.18	16.96	6.63	79.91	66.54	4.87	1.35	83.60	71.83	1.00	0.33
	MACL (Ours)	**85.82**	**75.16**	**4.85**	**1.91**	**88.96**	**80.12**	17.61	6.79	**82.93**	**70.84**	3.85	1.24	**84.13**	**72.61**	0.99	**0.32**
100 %	Scratch	89.92	81.69	4.59	2.12	90.11	82.01	15.71	6.17	91.52	84.36	2.45	0.61	87.00	77.00	0.81	0.27

datasets from T1-DUAL in/out phase and T2-SPIR; we use the T2-SPIR subset (20 images) with liver, kidneys, and spleen.
ROI-based Segmentation: **BraTS2018 dataset** [28] (39064 MRI slices) includes multi-modal MRI brain scans from 285 patients with tumor annotations and is used for both multi-organ and ROI-based MRI segmentation pre-training. **KiTS2019 dataset** [29] (32332 CT slices) provides 210 patients with kidney and tumor labels and is used for ROI-based CT segmentation pre-training. From **MSD dataset** [30], we select three tasks: **Heart** (Task02, MRI dataset, 20 patients labeled with left atrium), **Hippocampus** (Task04, MRI dataset, 260 patients labeled with anterior and posterior of hippocampus) and **Spleen** (Task09, CT dataset, 41 patients labeled with spleen). **ISIC2018 dataset** [31] includes 2594 dermoscopic images for skin lesion segmentation.

B. Implementation Details

We use our proposed MACL framework to pre-train a U-Net encoder with a two-block decoder on CHD, BraTS, and KiTS datasets in a fully unsupervised manner. The pre-trained model is then fine-tuned on eight downstream segmentation tasks using only 10% or 25% of labeled samples. The decoder is extended during fine-tuning to produce full-resolution outputs. Data pre-processing follows the protocol of PCL [4], with an 8:2 train-test split for all datasets. Experiments are conducted in PyTorch on NVIDIA A100 80G GPUs. For pixel-level contrastive loss, we found that using original images as spatial invariance groups yields better performance than non-spatial augmentations (e.g., brightness or contrast). Therefore, non-spatial augmentation is identity mapping, and spatial augmentations are translation, rotation and scale. During pre-training, we use the following hyperparameters: $\{\lambda_1, \lambda_2, \lambda_3\} = \{1.0, 0.5, 1.0\}$, temperature $\tau = 0.1$, SGD optimizer with an initial learning rate of 0.1, cosine-decayed to 0, 16 batches per GPU, and 100 training epochs. For fine-tuning, U-Net is trained with cross-entropy loss using the Adam optimizer, an initial learning rate of $5\times e^{-4}$ (cosine-decayed to $5\times e^{-6}$), batch size of 5 per GPU, and 100 epochs. We assess segmentation performance using four metrics: Dice Similarity Coefficient (DSC), Jaccard Coefficient (JC), 95th percentile Hausdorff Distance (HD95), and Average Surface Distance (ASD).

C. Comparisons with SOTAs

We compare MACL with training from scratch and 11 state-of-the-art CL methods. Notably, SimCLR, MoCo, PCL and DeSD are *symmetric one-stage* CL frameworks pre-training *only encoder*; GCL is a *symmetric two-stage* CL framework pre-training *encoder and decoder separately*; BYOL, SwAV, SimSiam and SimTriplet are *asymmetric one-stage* CL frameworks pre-training *only encoder*; DiRA is a *two-stage hybrid* SSL framework which has *symmetric* CL pre-training *only encoder*. However, our proposed MACL is an **asymmetric one-stage** CL framework pre-training **both encoder and decoder** together. All models use the same backbone and dataset splits. Pre-training and fine-tuning follow a consistent setup: CHD (CT) → ACDC/MMWHS, BraTS (MRI) → HVSMR/CHAOS for multi-organ segmentation; KiTS (CT) → Spleen/ISIC, BraTS → Heart/Hippocampus for ROI segmentation.
Quantitative results: Tables I and II demonstrate that our proposed MACL framework consistently outperforms existing SOTA methods across various metrics and annotation ratios, owing to its asymmetric design and multi-level contrastive learning strategy that jointly pre-trains both encoder and decoder. Specifically, (1) **With 10% annotated data**, our MACL yields substantial improvements across all datasets. For instance, it achieves +1.72% DSC on ACDC compared to PCL, +7.87% DSC and +8.61% JC on MMWHS over GCL, and +7.53% DSC and +10.0% JC on Spleen. MACL also achieves the lowest HD95 on multiple datasets (e.g.,

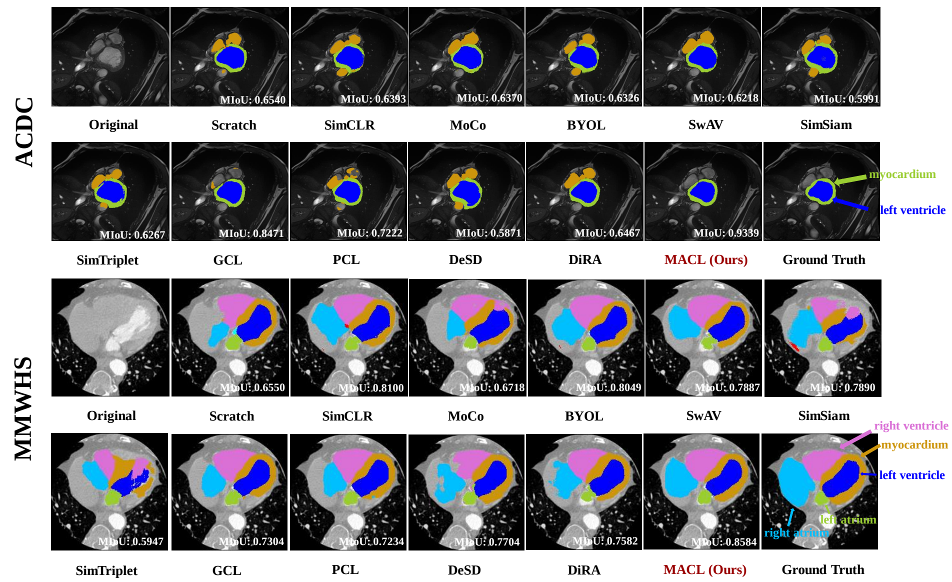

Fig. 3. Visualization of multi-organ segmentation results on ACDC and MMWHS datasets. Our proposed MACL achieves a superior performance with a higher MIoU and more precise predictions of substructures across other 11 methods.

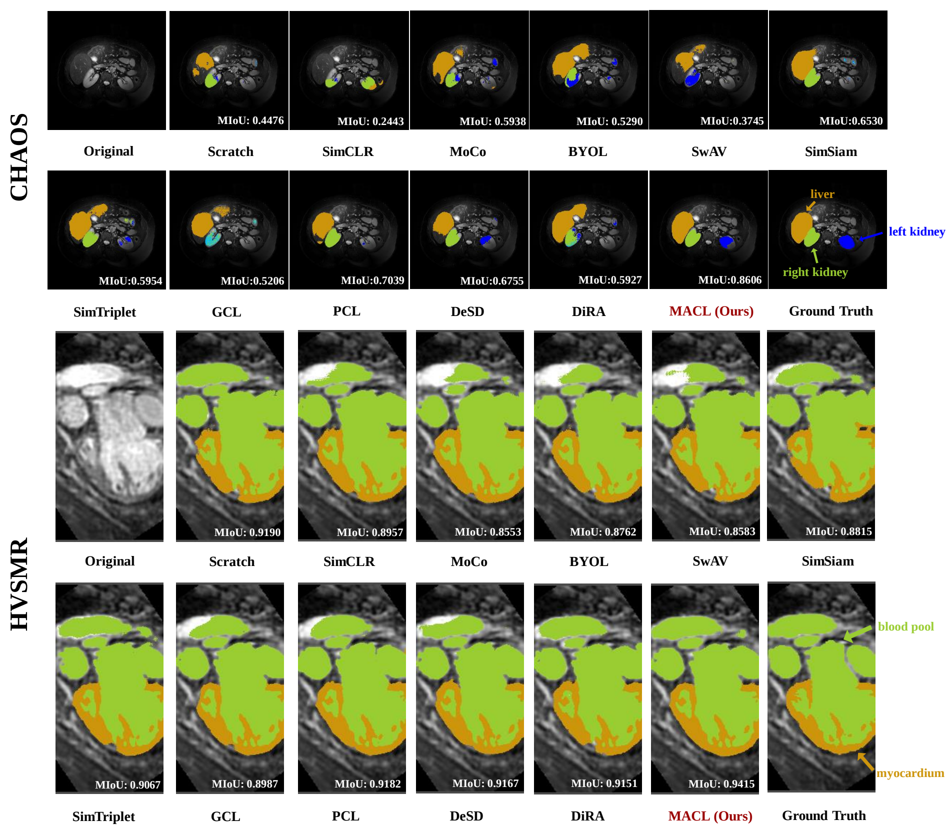

Fig. 4. Visualization of multi-organ segmentation results on CHAOS and HVSMR datasets. Our proposed MACL achieves a superior performance with a higher MIoU and more precise predictions of substructures across other 11 methods.

5.23 on ACDC, 5.91 on MMWHS, 6.93 on Spleen, 22.38 on HVSMR, and 21.53 on ISIC). (2) **With 25% annotated data**, although performance gains narrow due to the stronger supervision, MACL still maintains consistent advantages. For example, it achieves +1.89% DSC and +2.88% JC over GCL on MMWHS, and +1.45% DSC and +3.42% JC on Spleen. MACL achieves top performance across all metrics on the four multi-organ datasets (e.g., 83.68% DSC, 72.47% JC, 17.03 HD95, and 4.14 ASD on HVSMR). (3) Moreover, our MACL with only 25% annotations approaches the performance of fully-supervised models trained on 100% labels, *e.g.*, on ACDC, MACL achieves 89.64% DSC vs. 92.07% and 1.20 vs.

TABLE III
THE ABLATION STUDIES FOR EACH COMPONENT OF OUR PROPOSED METHOD.

Settings					ACDC (25% label)				MMWHS (25% label)			
Framework	$\mathcal{L}_g$	*decoder*	$\mathcal{L}_{ER}$	$\mathcal{L}_d$	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓
PCL	✓				88.97	80.33	4.17	1.43	86.65	76.90	4.23	1.28
PCL + decoder	✓	✓			89.19	80.73	3.92	1.33	88.19	79.15	4.10	1.19
MACL (image-level)	✓	✓	✓		89.29	80.82	4.08	1.30	88.96	80.51	3.74	**0.94**
MACL (image/pixel-level)	✓	✓	✓	✓	**89.64**	**81.40**	**3.59**	**1.20**	**89.90**	**81.92**	**3.68**	1.01

TABLE IV
THE ABLATION STUDIES OF NUMBER OF BLOCKS OF THE DECODER. $\boldsymbol{\lambda}$ DENOTES SCALE FACTOR USED IN DOWNSAMPLING.

	ACDC (10% label)				MMWHS (10% label)			
Number of Blocks	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓
0	84.91	74.16	6.54	2.65	65.98	52.32	8.07	3.16
1 ($\lambda = 0.5$)	85.79	75.44	7.89	2.59	71.51	58.45	7.84	2.58
2 ($\lambda = 0.25$)	**86.63**	**76.72**	**5.23**	**1.73**	**79.28**	**67.19**	**5.91**	1.98
3 ($\lambda = 0.125$)	83.61	72.17	9.66	3.22	73.33	60.42	7.29	2.22
4 ($\lambda = 0.0625$)	85.70	75.34	8.24	2.86	75.57	62.79	6.39	**1.96**

TABLE V
THE SENSITIVITY STUDIES OF $\boldsymbol{\lambda_2}$ ($\boldsymbol{\mathcal{L}_d}$) AND $\boldsymbol{\lambda_3}$ ($\boldsymbol{\mathcal{L}_{ER}}$) ON 4 MULTI-ORGAN DATASETS WITH 10% ANNOTATIONS.

	Settings		ACDC (100 patients)				MMWHS (20 patients)				HVSMR (10 patients)				CHAOS (20 patients)			
Label	λ_2 ($\mathcal{L}_d$)	λ_3 ($\mathcal{L}_{ER}$)	DSC (%)↑	JC (%)↑	HD95↓	ASD↓	DSC (%)↑	JC (%)↑	HD95↓	ASD↓	DSC (%)↑	JC (%)↑	HD95↓	ASD↓	DSC (%)↑	JC (%)↑	HD95↓	ASD↓
	0.25	0.25	85.26	74.62	6.33	2.38	76.49	63.60	6.98	2.29	76.92	63.53	24.08	6.75	66.14	**51.81**	**14.95**	**5.81**
	0.25	0.5	85.90	75.62	5.84	2.12	75.50	62.83	6.14	**1.75**	77.30	64.19	24.69	6.91	62.39	47.69	16.69	6.59
	0.25	0.75	84.70	73.82	7.15	2.56	76.76	64.53	6.11	1.80	76.19	62.89	22.99	6.03	60.22	45.08	18.16	7.04
	0.25	1.0	85.85	75.58	7.35	2.88	76.08	63.45	6.45	1.95	77.87	64.77	23.50	6.78	62.67	48.73	15.78	6.16
	0.5	0.25	84.79	73.97	6.84	2.46	75.24	62.54	7.00	2.54	75.70	62.29	27.61	7.43	61.39	46.63	17.48	6.98
	0.5	0.5	86.13	75.93	5.45	2.08	73.79	61.37	6.82	2.12	76.65	63.50	24.65	6.77	62.73	48.21	17.24	6.62
	0.5	0.75	85.72	75.30	6.13	2.36	75.02	62.65	6.84	1.90	76.95	63.81	25.24	6.85	60.82	46.08	17.56	7.02
10%	**0.5**	**1.0**	**86.63**	**76.72**	5.23	**1.73**	**79.28**	**67.19**	**5.91**	1.98	**78.77**	**66.05**	22.38	6.25	65.09	51.12	18.95	7.40
	0.75	0.25	85.17	74.51	**5.11**	1.94	78.32	66.37	6.17	1.75	77.71	64.63	25.64	7.47	62.71	48.11	16.51	6.24
	0.75	0.5	85.38	74.81	7.45	2.55	72.63	59.48	6.85	2.14	77.62	64.67	24.36	6.39	**66.36**	51.55	17.03	6.42
	0.75	0.75	85.48	74.91	6.92	2.42	74.48	61.97	6.61	1.90	77.77	64.86	22.93	6.30	65.99	51.11	17.07	7.02
	0.75	1.0	84.43	73.36	8.02	3.41	77.69	65.31	6.29	1.83	76.80	63.56	25.42	6.46	65.72	51.36	16.85	6.62
	1.0	0.25	85.61	75.10	6.66	2.38	73.38	60.59	6.83	2.27	77.88	64.90	22.74	**5.60**	62.30	47.91	17.48	6.58
	1.0	0.5	84.76	73.87	7.16	2.64	76.01	63.20	6.79	2.13	78.47	65.56	23.54	6.84	65.69	51.09	15.17	6.10
	1.0	0.75	85.33	74.67	5.89	2.21	77.46	64.95	5.92	1.78	77.87	64.90	**21.46**	6.35	64.41	49.95	17.41	7.00
	1.0	1.0	86.05	75.75	6.19	2.40	77.64	65.07	6.34	1.83	77.99	64.93	22.70	6.45	61.47	48.15	16.95	6.80

1.07 ASD; on HVSMR, 83.68% vs. 84.95% DSC; on ISIC, 88.96% vs. 90.11% DSC and 6.79 vs. 6.17 ASD.

Visualization results: Our comprehensive visualization analysis across multi-organ and ROI-based segmentation tasks (Figures 3-5) consistently demonstrates MACL's superior performance. In multi-organ segmentation (ACDC, MMWHS, CHAOS, HVSMR), MACL achieves higher MIoU with more precise substructure delineation, accurately segmenting challenging regions like the right ventricle in ACDC and right atrium in MMWHS where all 11 baseline methods show errors or imperfections. Similarly, for ROI-based segmentation (Spleen, Heart, ISIC), MACL produces more complete and accurate predictions, notably achieving MIoU 0.9559 for left atrium segmentation in the Heart dataset while methods like DeSD and DiRA exhibit significant missing regions and SimSiam/SimTriplet fail entirely. These visual results confirm MACL's enhanced capability in handling both complex multi-organ structures and focused ROI segmentation.

D. Discussion

Ablation of key components of MACL: We conducted ablation experiments to evaluate the key components in MACL, including the asymmetric decoder, equivariant regularization $\mathcal{L}_{ER}$ and dense contrastive loss $\mathcal{L}_d$. As shown in Table III, each component enhances performance, with the decoder and $\mathcal{L}_d$ yielding the most notable improvements. *e.g.*, adding the decoder results in 1.54% DSC / 2.25% JC gain on MMWHS, while $\mathcal{L}_d$ brings 0.94% DSC / 1.41% JC gains on MMWHS.

Sensitivity analysis of λ_2 and λ_3: We have conducted a sensitivity analysis as detailed in Table V. This table presents a systematic grid search of hyperparameters of λ_2 (for $\mathcal{L}_d$) and λ_3 (for $\mathcal{L}_{ER}$) ranging from 0.25 to 1.0 (we set λ_1 as 1.0) across 4 multi-organ datasets with 10% annotations. According to Table V, the combination ($\lambda_1 = 1.0, \lambda_2 = 0.5, \lambda_3 = 1.0$) consistently yields robust and superior performance across these diverse datasets. Specifically, it achieves the highest performance: 86.63% DSC on ACDC, 79.28% DSC on MMWHS and 78.77% DSC on HVSMR, and also a comparable performance on CHAOS with 10% annotations. Therefore, we set our hyperparameters as $\{\lambda_1, \lambda_2, \lambda_3\} = \{1.0, 0.5, 1.0\}$.

Decoder depth ablation: We evaluated the impact of decoder depth ($N = 0$–4) on performance. As shown in Table IV, all decoder-equipped models outperform the version without a decoder. The two-block decoder achieves the best overall results across most metrics (except ASD on MMWHS) and is adopted as the default. Deeper decoders (3 or 4 blocks)

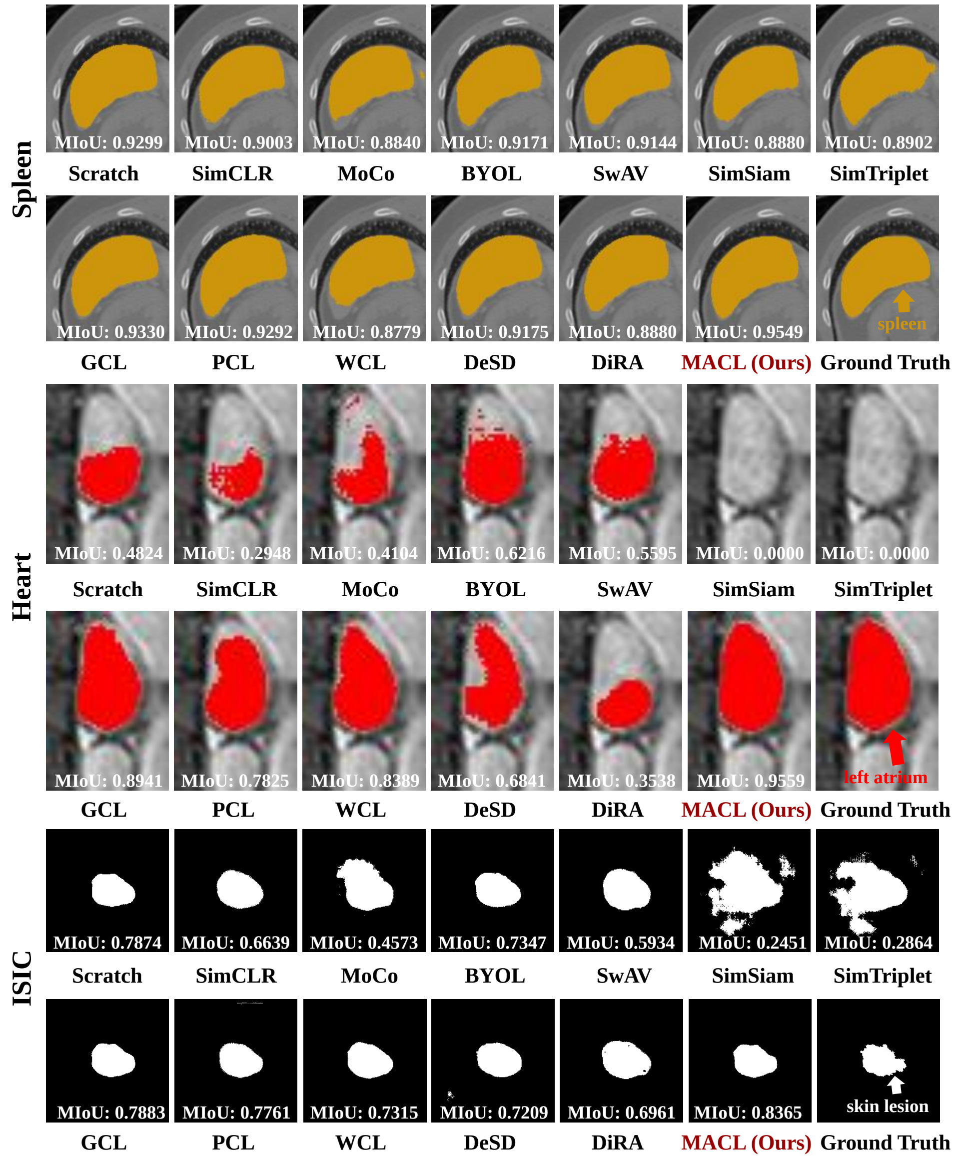

Fig. 5. Visualization of ROI-based segmentation results on Spleen, Heart and ISIC datasets.

introduce excessive downsampling, reducing input resolution and harming segmentation accuracy.

Different connection modes of encoders: Considering there exists different connection modes of encoders in these SOTA methods, such as parameter-shared encoders in SimCLR and SimSiam, and encoder with EMA in MoCo and BYOL. We explore the performance of different connection modes of the encoder between two branches. From Table VI, we can conclude that the parameter-shared mode matters a lot especially in CL with negative sample pairs. This is because using independent encoders or EMA-based encoders leads to misaligned feature spaces, which weakens contrastive supervision – especially when explicit negative pairs and dense pixel-level contrasts are involved. In contrast, parameter-shared encoders ensure consistent representations across branches, allowing multi-level contrastive losses to jointly and stably optimize the encoder. Moreover, we further explore the performance between one-stage MACL (pre-train encoder and decoder simultaneously) and two-stage MACL (pre-train encoder and decoder separately). From Table VI, we can see that our MACL in one-stage setting performs much better than MACL in two stage setting (The connection mode of encoders in Stage two of Two-stage MACL is parameter-shared).

Generalization to U-Net variants: To demonstrate the universality of MACL, we tested it on various U-Net variants, including AttUNet [10] (decoder with attention blocks), UCTransNet [11] (Transformer complement to CNN-based U-Net), RollingUNet [12] (incorporate MLP into U-Net) and UKAN [13] (integrate Kolmogorov-Arnold Networks into U-Net) for comparison with the basic U-Net [9]. Results in Figure 6 show that MACL consistently improves performance (DSC and JC) on both ACDC and MMWHS, confirming its strong generalization across different U-Net-based architectures.

Ablation studies of λ_2: According to Table III, we conclude that pixel-level dense CL loss $\mathcal{L}_d$ matters more than equiv-

TABLE VI
COMPARISON RESULTS OF DIFFERENT CONNECTION MODES OF ENCODERS AND DIFFERENT STAGES OF MACL.

Labeled%	Settings: Stages	Settings: Framework	ACDC DSC (%) ↑	ACDC HD95 ↓	MMWHS DSC (%) ↑	MMWHS HD95 ↓
10 %	One Stage	two independent encoders	82.20	6.24	63.17	9.82
		encoders with EMA	79.75	10.96	59.61	10.84
		two parameter-shared encoders	**86.63**	**5.23**	**79.28**	**5.91**
	Two Stages	stage one **w** downsample	81.15	6.99	60.13	10.53
		stage one **w/o** downsample	81.46	7.13	59.73	11.02
25 %	One Stage	two independent encoders	89.09	4.73	86.68	4.14
		encoders with EMA	88.24	4.94	86.50	4.49
		two parameter-shared encoders	**89.64**	**3.59**	**89.90**	**3.68**
	Two Stages	stage one **w** downsample	88.19	4.65	85.38	4.24
		stage one **w/o** downsample	89.28	3.80	84.60	5.01

TABLE VII
QUANTITATIVE RESULTS OF OUR PROPOSED MACL AGAINST MAEVIT ON 4 MULTI-ORGAN AND 4 ROI-BASED SEGMENTATION DATASETS UNDER 10% AND 25% ANNOTATIONS.

Labeled	Methods	ACDC (100 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD↓	MMWHS (20 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD↓	HVSMR (10 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD↓	CHAOS (20 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD↓
10 %	MAEViT [32]	77.58	64.00	6.56	2.70	70.75	56.60	7.85	2.44	67.35	52.56	28.25	6.85	65.08	49.56	**11.68**	**4.81**
	MACL (Ours)	**86.63**	**76.72**	**5.23**	**1.73**	**79.28**	**67.19**	**5.91**	**1.98**	**78.77**	**66.05**	**22.38**	**6.25**	**65.09**	**51.12**	18.95	7.40
25 %	MAEViT [32]	83.27	71.73	5.23	1.96	86.98	77.34	**3.59**	1.04	75.15	61.13	27.25	6.29	79.30	66.22	**7.60**	**2.98**
	MACL (Ours)	**89.64**	**81.44**	**3.59**	**1.20**	**89.90**	**81.92**	3.68	**1.01**	**83.68**	**72.47**	**17.03**	**4.14**	**79.88**	**67.45**	8.07	3.05

Labeled	Method	Spleen (41 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD↓	ISIC (2594 images) DSC (%)↑	JC (%)↑	HD95↓	ASD↓	Heart (20 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD↓	Hippocampus (260 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD↓
10 %	MAEViT [32]	79.87	66.48	18.08	6.57	82.86	70.74	26.94	10.89	48.61	32.11	5.81	2.60	75.93	61.21	**1.28**	0.49
	MACL (Ours)	**80.93**	**67.97**	**6.93**	**2.92**	**87.45**	**77.70**	**21.53**	**8.91**	**79.37**	**65.79**	**5.40**	**1.99**	**78.33**	**64.38**	1.31	**0.46**
25 %	MAEViT [32]	**86.58**	**76.33**	9.99	2.60	84.13	72.60	28.29	11.95	58.82	41.67	5.56	2.24	79.39	65.83	1.08	0.38
	MACL (Ours)	85.82	75.16	**4.85**	**1.91**	**88.96**	**80.12**	**17.61**	**6.79**	**82.93**	**70.84**	**3.85**	**1.24**	**84.13**	**72.61**	**0.99**	**0.32**

TABLE VIII
COMPUTATIONAL EFFICIENCY ANALYSIS BETWEEN OUR PROPOSED MACL AND OTHER BASELINE METHODS ON CHD DATASET.

Method	Parameters (M)	Avg Epoch Time (s)	Avg Iter Time (ms)	Peak Memory (MB)	Total Time (min)	Total Time (h)
SimCLR [18]	5.24	242.92	190.68	9473.69	404.87	6.75
MoCo [17]	6.29	118.67	66.32	2284.90	197.78	3.30
BYOL [20]	14.69	299.04	242.48	10965.65	498.40	8.31
SwAV [19]	9.02	328.53	270.16	23892.60	547.55	9.13
SimSiam [21]	5.37	237.84	181.03	9476.88	396.40	6.61
SimTriplet [7]	5.37	330.34	271.13	14100.94	550.57	9.18
GCL [8]	5.24	237.05	184.40	9474.82	395.08	6.58
PCL [4]	5.24	241.45	184.57	9474.82	402.42	6.71
DeSD [5]	36.76	347.53	279.27	12784.51	579.22	9.65
DiRA [6]	8.52	121.08	66.35	2390.54	201.80	3.36
MACL (Ours)	8.63	285.07	208.57	10915.44	475.12	7.92

TABLE IX
THE BREAKDOWN OF OUR MACL'S COMPUTATIONAL OVERHEAD.

Component	Avg Epoch Time (s)	Percentage (%)
Forward Pass	80.67	28.30
∟ Preprocessing	0.28	0.10
∟ Asymmetric CL Structure	80.39	28.20
∟ Parameter-shared Encoders	47.00	16.49
∟ Partial Decoder	28.76	10.09
∟ Projectors	4.63	1.62
$\mathcal{L}_{MLC}$ (Multi-level CL Strategy)	6.01	2.11
∟ $\mathcal{L}_g$ (Global Contrastive Learning)	0.63	0.22
∟ $\mathcal{L}_d$ (Dense Contrastive Learning)	0.45	0.16
∟ $\mathcal{L}_{ER}$ (Equivariant Regularization)	4.93	1.73
Other Overhead	198.39	69.59
Total	**285.07**	**100**

ariant regularization $\mathcal{L}_{ER}$, therefore we conduct a detailed ablation experiment of the weighted term λ_2 of $\mathcal{L}_d$. From Figure 7, we conclude that our proposed pixel-level dense CL loss $\mathcal{L}_d$ has a local optimum within the range of 0.1 to 0.9, where λ_2 = 0.5 achieves the best performance, which is also used as our baseline setting. Specifically, when λ_2 = 0.5, our proposed MACL achieves the best DSC (86.63% and 79.28%) and HD95 (5.23 and 5.91) on both ACDC and MMWHS with 10% annotations.

Feature visualization: To further investigate the representational differences, we also visualize the feature heatmaps on pretraining CHD dataset output from our proposed MACL framework and other baseline methods in Figure 8. Based on the visual evidence, we observe that the feature map from our MACL framework demonstrates the highest qualitative alignment with the Ground Truth mask. Its activations are the most focused, semantically precise, and spatially coherent with the target anatomical structure. This superior visualization provides direct visual evidence that MACL learns more discriminative and anatomically meaningful representations compared to other contrastive learning baselines, confirming the effectiveness of its overall design including asymmetric CL structure with a partial decoder and multi-level CL strategy.

Computational efficiency analysis: We have also conducted a detailed performance analysis between our MACL and other baseline methods. Our experiments are carried out on an

TABLE X
QUANTATIVE RESULTS OF OUR PROPOSED MACL AGAINST THE BASELINE METHOD PCL IN REAL-WORLD SCENARIOS, INCLUDING NOISY LABELS (BOUNDARY AMBIGUITY, LABEL FLIPPING, AND PARTIAL MISSING ANNOTATIONS) AND CLASS IMBALANCE.

Labeled	Settings	PCL		MACL	
		DSC (%)↑	JC (%)↑	DSC (%)↑	JC (%)↑
Noisy Annotations	boundary	**70.47**	61.04	70.19	**62.04**
	flip	66.58	56.17	**70.81**	**62.81**
	missing	57.39	46.99	**62.95**	**52.07**
Class Imbalance	ratio 0.1	19.09	14.30	**56.40**	**45.99**
	ratio 0.3	47.82	38.19	**71.60**	**62.73**
	ratio 0.5	64.41	52.57	**73.61**	**64.92**

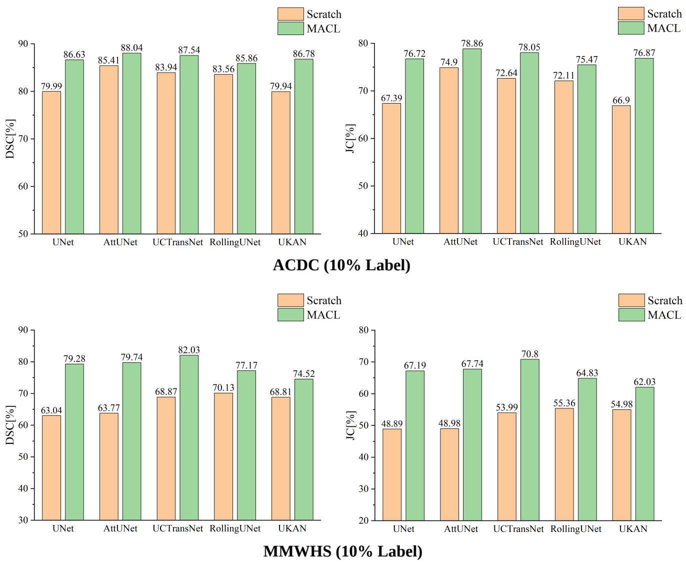

Fig. 6. Comparison results of different variants of UNet backbone on ACDC and MMWHS.

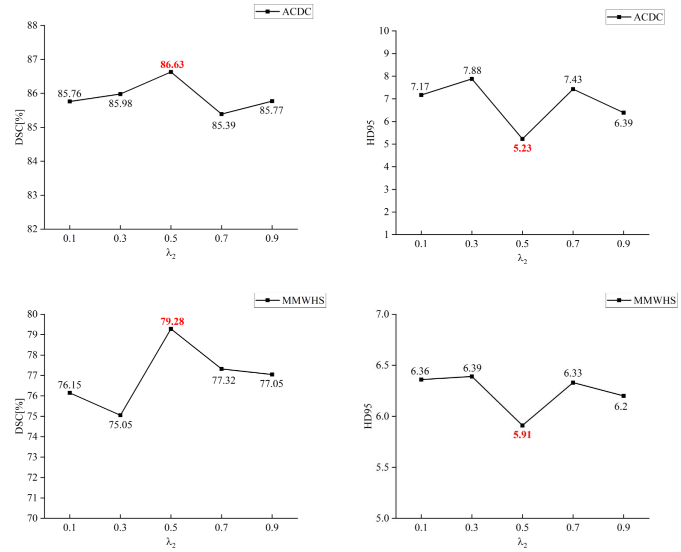

Fig. 7. Comparison results of different $\boldsymbol{\lambda_2}$ on ACDC and MMWHS with 10% annotations. And $\boldsymbol{\lambda_2} = \mathbf{0.5}$ achieves the best which is used as baseline setting.

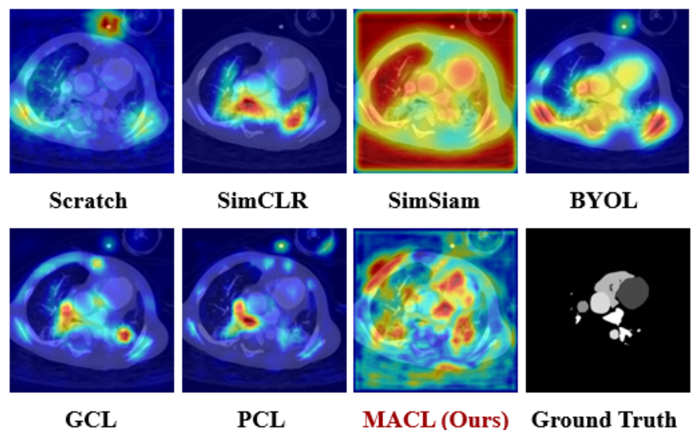

Fig. 8. Visualization of the feature heatmaps on CHD dataset between our proposed MACL and other baseline methods.

NVIDIA A100 80G GPU, using a batch size of 16 and 16 parallel workers (*num_workers=16*) to optimize data loading. As shown in Table VIII, our MACL framework demonstrates highly competitive efficiency. With a total pre-training time of 7.92 hours, it is faster than many SOTA methods (*e.g.*, SwAV, SimTriplet) and only moderately slower than the most efficient baselines (*e.g.*, SimCLR, MoCo). Its peak memory usage (∼10.92 GB) is comparable to GCL/PCL (∼9.25 GB). Although incorporating a partial decoder increases parameters to 8.63M, MACL still achieves an excellent performance-efficiency balance. Furthermore, Table IX provides a crucial breakdown of our MACL's computational overhead within an epoch, which says: (1) The Asymmetric CL Structure (Parameter-shared Encoders + Partial Decoder + Projectors) constitutes 28.20% (80.39s) of the per-epoch time. Within this, the novel Partial Decoder accounts for 10.09% (28.76s), representing a necessary and justified overhead for learning hierarchical, segmentation-aware features; (2) The Multi-level CL Strategy ($\mathcal{L}_{MLC}$) is remarkably lightweight, consuming only 2.11% (6.01s) of the total time, including $\mathcal{L}_g$ 0.22% (0.63s), $\mathcal{L}_d$ 0.16% (0.45s) and $\mathcal{L}_{ER}$ 1.73% (4.93s), demonstrating that its multi-level learning objective is achieved with minimal computational cost. In summary, the core innovations of our MACL – Asymmetric CL Structure and Multi-level CL Strategy – are both computationally lightweight and efficient.

MAEViT vs. MACL: We conducted a fair comparison between our method, MACL, and a MAE-based Vision Transformer (MAEViT) under identical settings. A MAE-based ViT-tiny model (*5.7M parameters*) was pre-trained from scratch on the same unlabeled datasets for 800 epochs. To ensure a rigorous evaluation for segmentation, we equipped the ViT encoder with a carefully designed FPN-SegFormer hybrid decoder to leverage multi-scale features. Despite these optimizations, MAEViT's performance in downstream segmentation tasks was significantly inferior to MACL, particularly under limited annotation settings (10% and 25% labeled data). As shown in Table VII, MACL achieved markedly higher Dice and Jaccard scores across 7 benchmark datasets except Spleen with 25% annotations. The performance gap was most pronounced on smaller datasets. For instance, on the MMWHS dataset with 25% labeled data, MACL achieves a Dice score of 82.93%, outperforming MAEViT by a substantial margin of +24.11%. This trend is consistent and underscores a critical

TABLE XI
COMPARATIVE STUDY BETWEEN 2D (OUR MACL) AND 3D SELF-SUPERVISED PRETRAINING METHODS.

Labeled	Method	Pre-training Dataset Size	ACDC (100 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD↓	CHAOS (20 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD↓	Heart (20 patients) DSC (%)↑	JC (%)↑	HD95↓	ASD ↓
10 %	VoCo [33] (3D)	1689 3D volumes (∼ 210k 2D slices)	77.19	63.58	31.53	11.26	52.24	38.63	13.35	4.78	71.32	55.42	15.11	4.76
	nnSSL [34] (3D)	113921 3D volumes	87.87	78.58	3.07	1.03	90.66	83.18	2.95	1.12	92.23	85.58	2.08	0.60
	MACL (2D)	56589 2D slices	86.63	76.72	5.23	1.73	65.09	51.12	18.95	7.40	79.37	65.79	5.40	1.99
25 %	VoCo [33] (3D)	1689 3D volumes (∼ 210k 2D slices)	85.24	74.36	6.46	2.41	71.29	56.34	8.14	2.71	76.30	61.68	13.71	4.45
	nnSSL [34] (3D)	113921 3D volumes	91.54	84.49	2.16	0.76	91.96	85.24	2.17	0.75	92.81	86.59	2.25	0.47
	MACL (2D)	56589 2D slices	89.64	81.40	3.59	1.20	79.88	67.45	8.07	3.05	82.93	70.84	3.85	1.24

TABLE XII
QUANTITATIVE RESULTS ON ACDC WITH 10% AND 20% ANNOTATIONS OF OUR MACL IN SEMI-SUPERVISED LEARNING SCENARIO.

Labeled%	Method	DSC (%) ↑	JC (%) ↑	HD95 ↓	ASD ↓
10 %	SL	77.66	66.40	2.41	0.59
	MT [35]	81.11	69.99	8.99	2.70
	UA-MT [36]	80.71	69.58	13.69	4.50
	SASSNet [37]	82.56	71.90	9.13	2.64
	DTC [38]	84.32	74.04	9.47	2.63
	URPC [39]	82.41	71.69	5.83	1.65
	CPS [40]	84.24	73.91	8.26	2.45
	MC-Net+ [41]	86.14	76.61	6.04	1.85
	BCP [42]	88.48	80.62	3.98	1.17
	PLGCL [43]	88.43	79.90	3.45	1.11
	MACL + BCP	**89.11**	**80.96**	**4.54**	**1.33**
20 %	SL	84.62	74.85	6.32	1.79
	MT [35]	85.46	75.89	8.02	2.39
	UA-MT [36]	85.16	75.49	5.91	1.79
	SASSNet [37]	86.45	77.20	6.63	1.98
	DTC [38]	87.10	78.15	6.76	1.99
	URPC [39]	85.44	76.36	5.93	1.70
	CPS [40]	86.85	77.96	5.48	1.64
	MC-Net+ [41]	87.10	78.21	5.04	1.56
	BCP [42]	89.52	81.62	3.69	1.03
	PLGCL [43]	88.48	80.19	4.06	1.19
	MACL + BCP	**89.76**	**81.93**	**3.53**	**1.08**
100 %	SL (upper bound)	91.53	84.76	2.41	0.59

weakness of the MAE approach in data-scarce environments. The main reason of the performance difference is due to MAEViT's tendency to overfit on small medical datasets, a well-documented challenge for Vision Transformers. This observation aligns directly with the established literature. As discussed in MDViT [44], *"due to the lack of inductive biases, such as weight sharing and locality, ViTs are more data-hungry than CNNs"*. The MAE framework, built upon a ViT architecture, inherently inherits this vulnerability. Its pixel-level reconstruction objective, while effective for learning detailed textures, may not prioritize learning the high-level, discriminative semantic features that are most crucial for segmentation, especially when fine tuning data is limited.

Performance in real-world scenarios: Our robustness evaluation demonstrates MACL's strong performance in clinically challenging scenarios. Under noisy annotation conditions (10% ACDC labels with simulated boundary ambiguities, label flipping, and missing annotations), MACL achieved superior Dice scores of 70.81% (flip) and 62.95% (missing), outperforming PCL by 4.23% and 5.56% respectively. In class imbalance tests (25% ACDC annotations with progressive retention ratios across severity levels), MACL maintained robust performance of 71.60% (ratio=0.3) and 73.61% (ratio=0.5), representing substantial improvements of 49.7% and 14.3% over PCL. This robustness stems from MACL's multi-level asymmetric contrastive learning framework, which integrates feature-level, image-level, and pixel-level representations through simultaneous encoder-decoder pre-training within an asymmetric structure, enabling more resilient learning from imperfect medical data.

Comparison with 3D self-supervised pre-training methods: We further extend our comparison to include 3D self-supervised learning methods, specifically VoCo [33] and nnSSL [34]. Following the original implementations, we load pre-trained model weights and fine-tune them under limited supervision conditions (10% / 25% labels). VoCo is fine-tuned for 1000 epochs, while nnSSL – pre-trained on an extremely large dataset—is fine-tuned for 100 epochs using nnUNet framework parameters. (i) According to Table XI, the performance gap among methods is strongly influenced by the scale of their pre-training data. VoCo is pre-trained on 1.6k 3D volumes (210k 2D slices), while nnSSL leverages a massive dataset of 113,921 3D volumes (OpenMind). In comparison, MACL uses 56,589 2D slices (CHD: 17,525 slices for ACDC/CHAOS; BraTS: 39,064 slices for Heart). As shown in Table XI, MACL consistently outperforms VoCo across all datasets (*e.g.*, +9.44% DSC on ACDC with 10% labels), demonstrating the effectiveness of our approach. (ii) Despite nnSSL's superior performance attributable to its extreme-scale pre-training, MACL demonstrates remarkable data efficiency. On ACDC with only 10% labeled data, MACL achieves a DSC of 86.63%, which is highly competitive with nnSSL's result of 87.87%, despite being pre-trained on only 56,589 2D slices—a small fraction of nnSSL's pre-training corpus. With 25% labeled data, MACL (DSC: 89.64%) narrows the gap further, approaching nnSSL (91.54%) while relying on far less pre-training data.

Semi-supervised learning scenario: Furthermore, we have evaluated our proposed MACL in the semi-supervised learning scenario. We utilize our MACL framework to pre-train the UNet encoder as the initialization for the semi-supervised method BCP [42] (CVPR 2023). As shown in Table XII, our MACL + BCP achieves superior performance compared to both the original BCP and other semi-supervised baseline methods. Specifically, with 10% annotations, MACL + BCP attains a DSC of 89.11% and a JC of 80.96%, outperforming both BCP (88.48% DSC, 80.62% JC) and PLGCL (88.43% DSC, 79.90% JC). At the 20% annotation level, the advantage is even clearer: MACL + BCP achieves 89.76% DSC and 81.93% JC, which surpasses the results of both BCP (89.52% DSC, 81.62% JC) and PLGCL (88.48% DSC, 80.19% JC). These results affirm that our MACL can serve as a highly effective pre-training strategy and provide a strong initialization for semi-supervised learning.

Limitation: (1) While our dense contrastive learning strategy effectively captures local semantics, it constructs positive pairs based on positional correspondence, which may not fully leverage the inherent symmetry of certain anatomical structures (e.g., left and right brain hemispheres). In future work, we plan to explore methods for incorporating anatomical prior knowledge, such as explicitly constructing positive pairs between symmetrical pixels, to further enhance the segmentation accuracy for symmetrical organs. (2) Despite the robust performance achieved with fixed loss weights, a limitation of our current approach is its static weighting scheme, which may not optimally adapt to diverse datasets or evolving training dynamics. In future work, we plan to investigate adaptive weighting mechanisms, such as those guided by Gaussian probabilistic models or principles of residual decay balancing. These strategies would dynamically adjust the weights of $\mathcal{L}_g$, $\mathcal{L}_d$ and $\mathcal{L}_{ER}$ based on real-time training signals (*e.g.*, gradient magnitudes), enhancing model flexibility across different data distributions and annotation scenarios, while also reducing the reliance on extensive manual hyperparameter tuning.

V. Conclusion

In this work, we propose a novel multi-level asymmetric CL framework named MACL for enhacing medical image segmentation. Specifically, we introduce an asymmetric CL structure and a multi-level CL strategy to ensure the encoder and decoder pre-train simultaneously and effectively. Experiments on multiple datasets indicate our MACL outperforms existing SOTA CL methods.

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VI. APPENDIX

A. Network Architecture

Our proposed MACL framework has three parts: an encoder $e(\cdot)$, a two-block decoder $d(\cdot)$ and projectors (image-level $g_{ins}(\cdot)$ and pixel-level $g_{pix}(\cdot)$). And during downstream finetuning, the two-block decoder $d(\cdot)$ will be extended to four-block to produce full-resolution outputs. The encoder and decoder are based on a UNet architecture: the encoder $e(\cdot)$ consists of 4 convolutional blocks and a center block. Each convolutional block consists of two "**Conv2d** $\rightarrow$ **BatchNorm2d** $\rightarrow$ **LeakyReLU**" structures followed by a 2×2 **Maxpooling** layer with stride 2. The kernel size of Conv2d is 3×3 with 1 zero padding. Similar to the encoder, the full decoder of UNet also consists of 4 convolutional blocks and a projection head. Each convolutional block consists of a **ConvTranspose2d** followed by two "**Conv2d** $\rightarrow$ **BatchNorm2d** $\rightarrow$ **LeakyReLU**" structures which is the same as those in the encoder. The projection head is a simple **Conv2d** structure which will project the fused features into a segmentation map. And our partial decoder is the first two blocks of the full decoder. The image-level projector $g_{ins}(\cdot)$ consists of an "**AdaptiveAvgPool2d** $\rightarrow$ **Flatten** $\rightarrow$ **Linear** $\rightarrow$ **ReLU** $\rightarrow$ **Linear**" structure. The pixel-level projector $g_{pix}(\cdot)$ consists of two "**Conv2d** $\rightarrow$ **BatchNorm2d** $\rightarrow$ **LeakyReLU**" structures. The kernel size of Conv2d is still 3×3 with 1 zero padding.

B. Pre-processing

Our dataset pre-processing employs a unified yet tailored approach to convert 3D medical volumes into 2D slices while preserving critical anatomical information. For cardiac imaging datasets, CHD is processed with a spatial resolution of $1.0 \times 1.0mm^2$ resampled to 512×512 dimensions, MMWHS uses $1.0 \times 1.0mm^2$ resampled to 256×256 and HVSMR applies $0.7 \times 0.7mm^2$ resampled to 352×352, all employing percentile-based intensity normalization $[x_1, x_{99}]$ with zero-padding to maintain structural integrity. The ACDC dataset follows nnUNet-based processing with spacing adjustment to (10, 1.25, 1.25) mm and z-score intensity normalization. BraTS2018 undergoes axial slicing with strategic cropping to 192×160 pixels, while abdominal datasets (KiTS, CHAOS) feature DICOM-to-PNG conversion with modality-specific intensity handling. Medical Segmentation Decathlon tasks (Heart, Hippocampus and Spleen) preserve native resolution without transformation, ensuring all processed data is systematically organized in numpy or PNG formats for compatibility with our 2D training framework while retaining dataset-specific characteristics essential for accurate segmentation.